# GRADE 10 – TEST 8

**Name: _____**
**Grade / Class: 10**

---

# ENGLISH

1. **Replace the underlined phrase with a precise single word:**
   **She completed the task without any help from others.**
2. **Identify the tense used in the sentence and justify your answer:**
   **By the end of the week, the engineers had repaired the bridge.**
3. **Combine the sentences using a relative pronoun:**
   **The scientist made a discovery. The discovery changed medical research.**
4. **Rewrite the sentence in indirect speech:**
   **He said, "I will submit the project tomorrow."**
5. **Correct the sentence by removing ambiguity:**
   **Visiting teachers can be confusing.**
6. **Explain how literature reflects social change, using general reasoning.**
7. **Write a descriptive paragraph about a place that represents peace to you.**
8. **Identify the tone of the statement and explain your choice:**
   **"Rules exist to guide us, not to imprison us."**
9. **Rewrite the sentence using a modal verb to express obligation:**
   **It is necessary for students to follow laboratory safety rules.**
10. **Write a short formal notice announcing a school-level science exhibition.**

---

# MATHEMATICS

1. **Factorize completely:**
   $x^2 - 11x + 30$
2. **Solve for z:**
   $3z - 4 = 2(z + 5)$
3. **Find the value of k if the point (k, −3) lies on the y-axis.**
4. **Calculate the distance between the points (1, 2) and (7, 10).**
5. **Express $7.2 \times 10^{-4}$ as a decimal number.**
6. **A rectangular solid has length 8 cm, width 5 cm, and height 4 cm. Find its volume.**
7. **Determine the value of tan 45° − sin 30°.**
8. **If the ratio of boys to girls in a class is 7:5 and there are 36 students, find the number of girls.**
9. **Find the perimeter of a semicircle with diameter 14 cm.**

10. A number is increased by 15 percent and becomes 92. What was the original number?

---

## PHYSICS

1. Define momentum and state its SI unit.
2. Explain why passengers lean backward when a bus suddenly starts moving.
3. Describe how energy is transformed in a hydroelectric power station.
4. State the relationship between voltage, current, and resistance.
5. Why does light bend when it passes from air into glass?
6. Explain the difference between scalar and vector quantities with examples.
7. Describe one practical application of electromagnetic induction.
8. What happens to resistance if the length of a wire is doubled?
9. Explain how sound travels through solids.
10. State one reason why mirrors are used in periscopes.

---

## CHEMISTRY

1. Define chemical equilibrium.
2. Explain why noble gases are generally unreactive.
3. Write the chemical formula of washing soda and state one use.
4. Differentiate between homogeneous and heterogeneous mixtures.
5. Explain the role of oxygen in combustion reactions.
6. What is meant by a neutralization reaction?
7. Describe one industrial use of ammonia.
8. Why are alloys preferred over pure metals for manufacturing tools?
9. Define pH and state its importance in daily life.
10. Explain how temperature affects the rate of a chemical reaction.

---

## BIOLOGY

1. Define homeostasis and explain its importance for survival.
2. Describe the function of red blood cells.
3. Explain how vaccination helps prevent diseases.
4. What is photosynthesis? State two conditions required for it.
5. Differentiate between arteries and veins.
6. Explain the role of enzymes in digestion.
7. Define ecosystem and name its two main components.

8. **Describe how roots absorb water from the soil.**
9. **What is heredity? Explain its significance.**
10. **Explain one harmful effect of environmental pollution on living organisms.**

---

## COMPUTER SCIENCE / ICT

1. **Define operating system and state one of its functions.**
2. **Explain the difference between primary memory and secondary storage.**
3. **What is a database? Give one real-world example.**
4. **Describe the purpose of encryption.**
5. **Explain how cloud computing benefits users.**
6. **Define algorithm and explain why it is important in programming.**
7. **What is the function of a firewall?**
8. **Explain the difference between input devices and output devices.**
9. **Describe one ethical issue related to social media use.**
10. **What is machine learning? State one area where it is applied.**

---

## ISLAMIYAT

1. **Define Taqwa and explain its impact on personal character.**
2. **Describe the importance of honesty in Islamic teachings.**
3. **Explain the concept of accountability in the Hereafter.**
4. **What is the significance of Zakat in society?**
5. **Describe one lesson from the life of Prophet Muhammad (PBUH) related to patience.**
6. **Explain why justice is emphasized in Islam.**
7. **Define Halal and Haram with one example each.**
8. **Explain the importance of seeking knowledge in Islam.**
9. **Describe the rights of neighbors according to Islam.**
10. **Explain how Islam promotes peace and tolerance.**